

PROFESSIONAL SUMMARY

Analytical, delivery-focused **Solutions Design Engineer** and Data Specialist with deep expertise in enterprise data architectures, regulatory reporting flows, data lineage, and systems integration within tier-1 banking. Currently operating across Santander UK’s **Cloud Data Foundation (CDF)**, leading the technical design, data controls, interface mapping, and reconciliation frameworks supporting complex data-driven change programmes. Adept at acting as the principal technical interface between Regulatory Reporting, Finance, Risk, Operations, and Technology teams to translate complex regulatory and business requirements into robust, controlled, and audit-ready technology solutions. Hands-on mastery in advanced SQL, data lineage tracing, reconciliation design, testing governance, system process re-engineering, and GenAI-assisted delivery tools.

CORE COMPETENCIES & TECHNICAL SKILLS

Solution Design & Architecture: End-to-End Solution Lifecycle, Technical Interface Design, Data Flow Mapping, System Process Re-engineering, Integration Patterns.	Regulatory Reporting & Governance: CFO Regulatory Reporting, Data Lineage & Traceability, Financial Control, Auditability, Policy Governance.
Data Quality, Controls & Reconciliation: 3-Tier Reconciliation (Row Counts, Column Hashes, Balance Totals), Automated Control Frameworks, Lake Formation ABAC, Metadata Tagging.	Data Engineering & Analytics Stack: Advanced SQL (Recursive CTEs, Window Functions), AWS (S3, Athena, Lake Formation, Glue), SAS, Python.
Delivery & Stakeholder Management: IT Project Management, Multi-Workstream Matrix Coordination, RAID Management, Governance Packs, Third-Party Vendor Coordination.	Testing, Change & GenAI Adoption: System Integration Testing (SIT), UAT Certification, Change Management, AI-Assisted Solution Design, Documentation & Code Refactoring.

PROFESSIONAL EXPERIENCE

Data Engineer & Solutions Design Specialist – Santander UK Feb 2023 – Present | UK

Enterprise Services / Cloud Data Foundation

Solution Design	Regulatory Reporting	Data Controls & Lineage	AWS (Athena/S3/Lake Formation)	Advanced SQL	GenAI Tools
End-to-End Solution Design & Technical Delivery: Architect and document fit-for-purpose solution designs, technical integration approaches, and data flows mapping complex banking datasets into Santander UK reporting platforms, ensuring complete end-to-end traceability from source systems through to downstream reporting submissions. Data Integration & Migration Readiness: Lead technical design and data interface specifications for integrating Plotly and CDSW, product, and transactional records into Santander UK’s target reporting architecture; establish consolidation logic, transformation mappings, and migration readiness gates. Cross-Functional Stakeholder Interface: Act as the principal technical interface bridging CFO Regulatory Reporting, Finance, Risk, Technology squads, and Business Assurance; translate complex regulatory calculation rules and business logic into robust, executable technical specifications. Technical Controls, Lineage & Reconciliation: Design, implement, and own rigorous data controls covering 3-tier reconciliation (volume parity, MD5/SHA256 field-level hash validation, and financial balance totals), data quality assertions, and automated schema validation to guarantee 100% reporting integrity. Data Governance & Access Control Architecture: Engineer and enforce attribute-based access control (ABAC) policies and column-level metadata classification tags within AWS Lake Formation, safeguarding sensitive financial PII and ensuring strict compliance with GDPR and banking regulations. Testing Strategy & Certification Cycles: Define and drive comprehensive testing strategies covering System Integration Testing (SIT), User Acceptance Testing (UAT), and dress rehearsal certification; automate test data generation and reconciliation pipelines, eliminating manual verification overhead. Project Management, Dependency & RAID Governance: Actively manage delivery risks, inter-squad dependencies, and technical blockers across 15+ engineering teams; author senior governance packs, delivery scorecards, and executive dashboards in Power BI. GenAI Tooling & Delivery Efficiency: Pioneer the adoption of GenAI tools to accelerate solution design documentation, complex SQL schema refactoring, synthetic test dataset generation, and project milestone tracking, cutting design-to-delivery lead times by 30%.					

Financial Crime, Customer Risk & Financial Analytics

Process Re-engineering SAS SQL Risk Modeling Automated Controls Governance

- **System Process Re-Engineering:** Redesigned and automated legacy SAS and SQL data pipelines evaluating customer inflows/outflows, income shocks, and account risk behaviors, replacing manual batch processes with automated, monitored data pipelines.
- **Data Quality & Lineage Controls:** Formulated automated exception handling and data quality validation frameworks that audited customer transaction histories, ensuring complete audit trails and data lineage for regulatory compliance.
- **Complex Account & Entity Modeling:** Engineered recursive SQL queries (CTEs, multi-table joins, window partitions) to analyze transaction velocity, multi-account pass-through patterns, and entity relationships across millions of records.
- **Executive MI & Reporting Interfaces:** Built interactive operational control dashboards in Power BI and Looker, presenting clear KPI metrics, risk trends, and remediation progress to senior risk and finance committees.

ACADEMIC MENTORSHIP & LEADERSHIP

Mentor – MSc Data Science & AI – University of East London

2023 – 2025 | London, UK

- Mentored postgraduate students on advanced data engineering, solution architecture principles, production SQL optimization, data governance frameworks, and applying generative AI tools to software design and analytics workflows.

EDUCATION & PROFESSIONAL CERTIFICATIONS

MSc in Information Technology Security Nottingham Trent University Graduated with Distinction Specialized in Information Governance & System Security	BSc in Physics University of Lagos Core modules: Software Engineering, Database Systems & Architecture
AWS Certified Data Engineer – Associate Amazon Web Services (AWS) Validation of Cloud Data Pipelines, Lake Formation, Glue, Athena & Security	AWS Certified Solutions Architect – Associate Amazon Web Services (AWS) Cloud Architecture Design, Multi-Tier Systems, Resiliency & Integration Patterns